

Communication Topology Effects on Accuracy in Multi-Agent LLM Reasoning Systems

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Repository: <https://github.com/SafoanMiah/llm-multi-agent-reasoning>

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1. Abstract

Multi-agent debate has emerged as an effective strategy for improving reasoning in large language models (Du et al., 2023; Liang et al., 2024), yet the communication structure governing agent interaction is rarely treated as an experimental variable, with most implementations defaulting to fully connected topologies or relatively big models.

This project evaluates the effect of four different communication topologies: independent, fully connected, mediator, and chain. This study evaluates 200 problems from the GSM8K benchmark using four small language models: Gemma3 4B, Phi4-Mini, Llama3.2 3B, and Qwen2.5 3B. Majority voting was used to determine the final group answer after five rounds of iterative discussion. When each of the four agents reasoned independently, the group accuracy was 18.0%.

When the agents collaborated through the fully connected, chain, and mediator topologies, the accuracy ranged from 63.0% to 84.0%, respectively. The mediator topology resulted in the highest accuracy and at the lowest token cost per correct answer achieving 84.0% accuracy. Additionally, the mediator topology was the only topology where the majority vote (84.0%) resulted in higher accuracy than each individual agent, with the best individual agent reaching 83.0% accuracy. These accuracy gains were concentrated in rounds two and three, with diminishing returns beyond round three relative to token expenditure. Regardless of the number of correct answers received by the group, the agents remained overconfident with a mean self-reported confidence over 90% for all topologies.

Statistical validation via Cochran's Q test ($p < 0.001$) and McNemar's tests with Bonferroni correction confirmed that the observed topology differences were significant across most comparisons; bootstrap confidence intervals demonstrated the stability of the results.

Finally, a supplementary experiment evaluated the performance of small-model collaboration versus single-model self-refining. Phi4-Mini iterating over five self refinement rounds without peer feedback achieved 53.0% accuracy, while the four collaborating 3-4B models under the mediator topology reached 75.5% at round 2, a +22.5 percentage point gain that isolates the specific contribution of inter-model communication over repeated self-critique alone.

2. Introduction

The idea of organizing multiple model instances into cooperative systems that collectively reason via structured interactions has emerged as a way of addressing multi-step reasoning challenges; even if a model can perform well on many natural language processing (NLP) tasks, they will still be unable to provide reliable answers to problems that require coherent reasoning across multiple intermediate steps. In fact, capable models often provide confidently incorrect answers when they are confronted with arithmetic chains or compositional problems. There is an increasing amount of research focused on this issue rather than scaling individual models, instead researchers are focusing on developing collaborative systems consisting of several model instances working together through structured interaction.

A recent example of such collaborative reasoning methods was the introduction of the multi-agent debate paradigm proposed by Du et al. (2023). The multi-agent debate paradigm works by having multiple LLMs independently generate candidate answers and then iteratively revising those candidate answers based on what the reasoning from the other agents indicate. The use of multiple model instances has been shown to lead to better factual accuracy and better mathematical reasoning than relying on single-agent models. Further research extending the framework of the multi-agent debate paradigm included methods for promoting divergent thinking, regulating the level of agreement among agents, and utilizing architecturally disparate agent pools. Specifically, Hegazy (2024) demonstrated that a diverse collection of smaller-sized models could out-perform even GPT-4 on some mathematical benchmarks solely due to debate, which suggests that the configuration of how the agents are defined may be equally important as how large the agents are defined.

However, nearly all of the above-referenced studies have fixed the communication structure prior to testing, which most commonly is a fully-connected graph where all agents observe the responses from all other agents in each round. The communication structure is rarely treated as a variable that may impact the outcome of the reasoning processes. The limited amount of work that does examine topology as a variable indicates that it does appear to matter: Li et al. (2024) found that sparse communication topologies were able to achieve accuracy similar to, or in excess of, that achieved through fully connected debate, but at significantly reduced token costs. Additionally, Qian et al. (2025) reported that irregular graph topologies out-performed regular graph topologies when the number of agents increased beyond one thousand. Despite these findings, controlled comparative studies examining the effect of different communication structures remain relatively rare. Most of the studies referenced in this review compare only two conditions, utilize expensive frontier (top) models, or examine multiple architectural dimensions concurrently. Therefore, the ability to isolate the effect of topology as a standalone factor is constrained.

This study aims to fill this gap through a controlled examination of four communication topologies: independent, fully connected, mediator, and chain applied to GSM8K mathematical reasoning problems. The system utilizes four heterogeneous small language models (each containing

approximately 3-4B parameters) while holding all other factors constant: the same prompts, the same number of rounds (where possible and needed), and the same method for aggregating the decisions made by the agents. Through the isolation of topology as the sole experimental variable, the study seeks to answer three questions:

- Do multi-agent collaborations result in higher accuracy than independent reasoning, and if so, to what extent?
- How do different communication structures affect the behaviour of agents during the convergence process and the rate at which agents agree?
- What are the cost/accuracy trade-offs across topologies, and does a structure exist that achieves high accuracy at lower computational expense?

2.1 Structure of the Dissertation

The remainder of this dissertation is organised as follows. Section 3 reviews relevant literature and previous research on multi-agent debate, communication topology, and reasoning in language models. Section 4 details the experimental design, implementation, and evaluation approach. Section 5 looks at the results. Section 6 discusses future work and limitations. Section 7 summarises the conclusions.

2.2 Aims and Objectives

The aim is to investigate the effect of communication topology on reasoning accuracy and convergence behaviour in multi-agent LLM systems using small open-source models, with the following objectives;

1. Design and implement a multi-agent reasoning framework that supports four communication topologies: independent, fully connected, mediator, and chain.
2. Evaluate group accuracy across topologies on 200 GSM8K mathematical reasoning problems under all four topologies.
3. Analyse convergence dynamics and agent agreement rates across iterative rounds.
4. Assess cost-efficiency trade-offs trade-off associated with each communication topology by determining how many tokens are spent by the agents to obtain one correct answer.
5. Use Cochran's Q, McNemar's tests and bootstrap confidence intervals, to statistically validate observed accuracy differences.
6. Compare multi-agent collaboration against a single-model self-refine baseline to isolate the contribution of inter-agent communication.

3. Background

The literature reviewed in this chapter provides the foundation for the design and assessment of the research presented in this dissertation. It begins with the foundational work on multi-agent debate, then examines what is known about the role of communication structure, finally reviews the reasoning methods and assessment methods which are used as a basis to develop the research methodology.

3.1 Multi-Agent Debate for LLM Reasoning

The idea that different language model instances could collaborate to provide better reasoning together was formalised by Du et al. (2023), who described a collaborative process among agents that involved each independent agent generating a response to a question; however, once each of the agents had generated an answer, they would read the output from all of the other agents and begin to iteratively adjust their answers based on the input from the other agents. In every category (arithmetic, strategic thinking and factual knowledge) tested using the multi-agent debate process the collaborative debate method produced significantly better results than the single-agent baseline results, even in cases where all of the agents were initially incorrect, but collectively determined a correct answer through dialogue. Following the development of the multi-agent debate method more researchers have begun to develop alternative versions of the framework. Liang et al. (2024) developed an asymmetric version of the debate format that utilized two separate types of agents, debaters, and judges.

Liang et al. (2024) argued that utilizing this type of debate format, and allowing agents to be both divergent and convergent in their thinking, would help to avoid premature convergence.

Smit et al. (2024) tested the efficacy of several debate formats across medical and general-knowledge categories. The research conducted by Smit et al. (2024) found that the amount of agreement between the agents at the beginning of the debate process significantly impacts the overall level of accuracy achieved in the final answer.

Wang, Q. et al. (2024) offered a more cautionary perspective, showing that a single agent with a sufficiently strong prompt can sometimes match multi-agent discussion performance, suggesting that the benefits of collaboration are not guaranteed and may depend on how the system is configured.

One of the most consistent findings across all of the above research is that the diversity of the models being used is important. For example, Hegazy (2024) demonstrated that a collection of three medium capacity models with different architectures achieved higher results on the GSM8k benchmark than GPT-4 did when debating, achieving an accuracy rate of 91%. This result is particularly relevant to the present project, which similarly uses a diverse set of small models rather than multiple instances of a single large one.

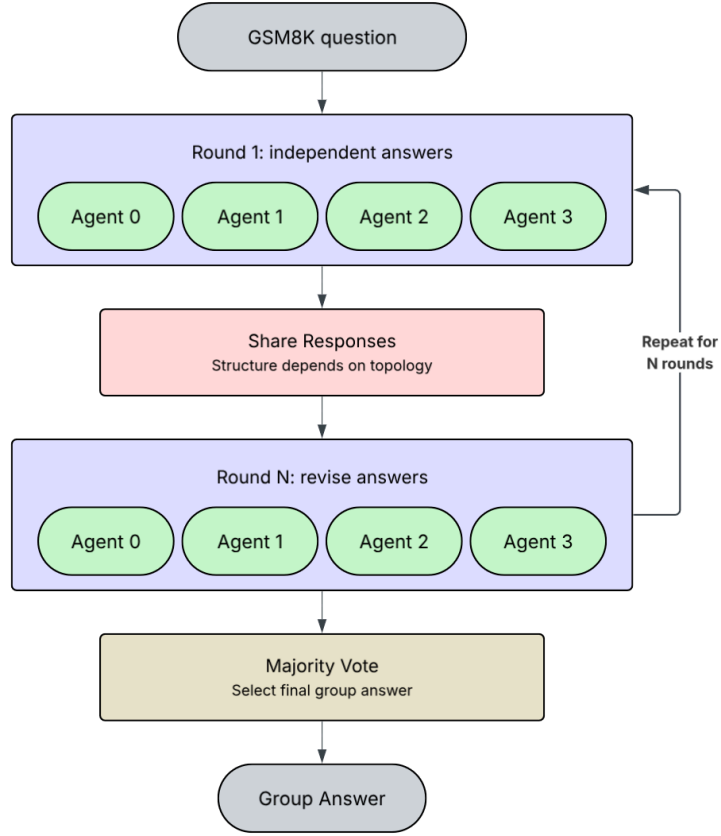


Figure 1

The iterative debate loop that's at the core of the system

3.2 Communication Topology as a Design Variable

Most multi-agent debate implementations employ a fully connected topology where every agent sees all other agents' responses at each round. The selection of this topology is generally not supported and is simply used as a default framework. There have been many studies recently that suggest the structure of communication has an impact on reasoning outcomes.

Li et al. (2024) completed direct comparisons evaluating fully connected vs. Neighbor connected (sparse) topologies on GSM8K and several other benchmarks using GPT and Mistral models, they also discovered that sparse topologies achieved comparable or better accuracy than fully connected networks while reducing token cost by up to ~44%. As multi-agent debate is already computationally expensive, this encourages the inclusion of the chain network topology in the present study which represents the sparsest possible connected network.

At a larger scale, Qian et al. (2025) introduced MACNET, a framework that organizes agents into directed acyclic graphs with various network topologies including chains, trees and random graphs. Their experiments across both reasoning and code generation tasks revealed two notable results: first, irregular (random) network topologies often performed better than regular network

topologies such as mesh or layered structures; second, performance follows a logistic pattern as the number of agents increases, a phenomenon they refer to as "collaborative scaling law". They attribute the advantage of irregular network topologies to small world properties, shorter average path lengths and higher clustering coefficients that improve information flow without the overhead of dense communication.

Zhang et al. (2025) took a complementary approach with G-Designer, proposing that the communication network topology itself should be learned rather than fixed. Zhang et al. (2025) used graph neural networks to architect task-specific network topologies and demonstrated that optimized structures perform better than both fixed fully connected and manually designed alternatives. Although Zhang et al. (2025) required a training phase beyond the scope of this project, their work reinforces the central premise: that topology is not a neutral implementation choice but an active design variable.

Another related issue is susceptibility of multi-agent systems to conformity effects. Estornell and Liu (n.d.) provided a theoretical framework for multi-LLM debate and identified a "tyranny of the majority" effect, in which agents tend to adopt the majority answer regardless of its correctness. This echo-chamber dynamic is similar to groupthink in human organizations and presents a problematic situation when shared training data causes multiple models to share misconceptions. Their proposed interventions, diversity pruning and misconception refutation, suggest that the communication network structure may either amplify or mitigate such effects. Chuang et al. (2024) observed a similar phenomenon in simulated LLM opinion dynamics, finding that agents interacting over communication networks tend to converge toward consensus and that rate and direction of convergence are influenced by network structure and inherent model biases consistent with earlier theoretical work on consensus dynamics in networked multi-agent systems (Chebotarev, 2010).

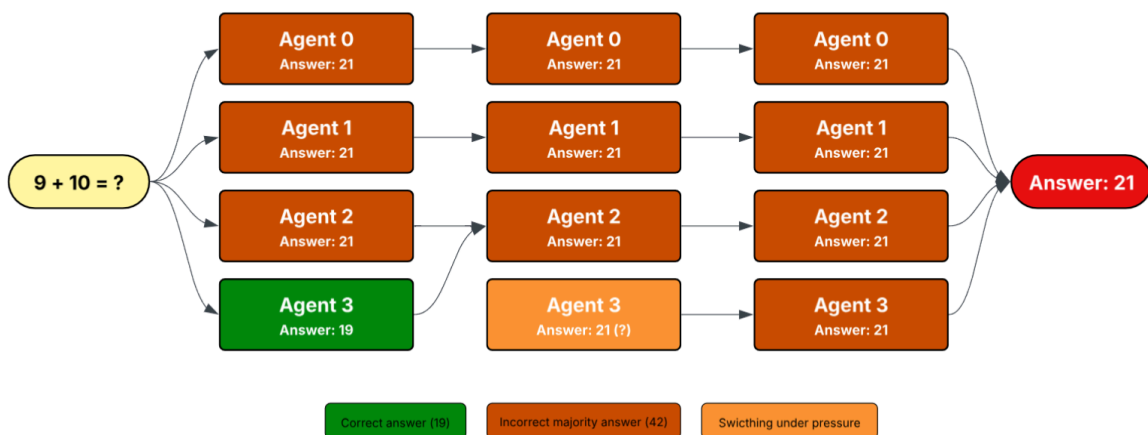


Figure 2

"Tyranny of the majority" Agent 3 held the correct answer but adopted the majority's incorrect answer after debate

3.3 Reasoning Techniques and Self-Correction

The methodological strategies employed by both of the agents were based upon a number of foundational approaches. One such example is chain-of-thought (CoT) prompting, which was shown by Wei et al. (2023) to significantly enhance the performance of language models on arithmetic and commonsense questions when they were prompted to generate intermediate reasoning steps. It is this CoT strategy that formed the basis of the structured output format in our multi-agent system; in addition to producing an answer, agents are required to provide step-by-step justifications for the answer produced.

Wang et al. (2023) expanded upon this strategy with self-consistency, showing that selecting the most common answer, through majority voting, among multiple reasoning paths sampled from a single model results in significant improvements in accuracy, up to +17.9% on GSM8K. Although the concept of self-consistency and our independent topology baseline are similar in that both use parallel reasoning and aggregation through vote, our agents are distinct models, as opposed to being samples from the same model.

A natural next question is whether a single model can better itself using self-feedback, i.e., iterating through an internal loop of self-refining without needing separate agents. Madaan et al. (2023) showed that Self-Refine could be used to refine a model's reasoning through iteration by having a model produce an initial answer, critique it, and then produce a revised answer. However, Huang et al. (2024) questioned the generalizability of this approach, showing that LLMs cannot reliably correct their own reasoning absent external feedback. They showed that self-correction would decrease performance on GSM8K when there was no external signal indicating that the model had produced an incorrect answer. Given the findings of Huang et al. (2024) regarding the reliability of self-correction absent external signals, we believe that feedback from other agents in a multi-agent system would be important, as opposed to being redundant.

3.4 Confidence Calibration in Language Models

The reported confidence levels by each agent in our framework is well understood, and the research of Kotelanski et al. (n.d.), as well as Yang et al. (2024) provides an empirical basis for this understanding. Specifically, the former study identified that large language model (LLM) confidence estimates are generally poorly calibrated; large language models tend to report higher confidence than they should when providing incorrect responses. The latter study found that all models studied were overly confident in their answers. In addition, both studies suggested that the extent to which models provide accurate confidence ratings will be influenced by the way in which they are prompted to estimate their uncertainty. We hypothesize that confidence levels provided by 3-4 billion parameter models in our experiments will likely be less reliable predictors of correctness than expected.

3.5 Test-Time Compute and Efficiency

The multi-agent debate model is equivalent to test/time compute scaling: Rather than investing additional training time and resources into the model, the model will use additional computational resources at inference time to produce higher-quality outputs. Snell et al. (2024) developed this trade-off conceptually and demonstrated that optimal allocation of test-time compute can be more efficient than allocating an equal amount of resources towards training or model parameter scaling. This work gives the theoretical basis for the cost-effectiveness evaluation in this study which evaluates how much tokens are required to reach a certain accuracy at each of the three architectures. Additionally, the Mixture-of-Agents (MoA) framework Wang, J. et al. (2024) also illustrates this approach, with iterative aggregation of responses from multiple models providing successively refined versions of the final response through additional "refinement" layers, similar to the mediator architecture proposed in this study, in which a summarizing agent summarizes the group's reasoning before individual agents revise their responses.

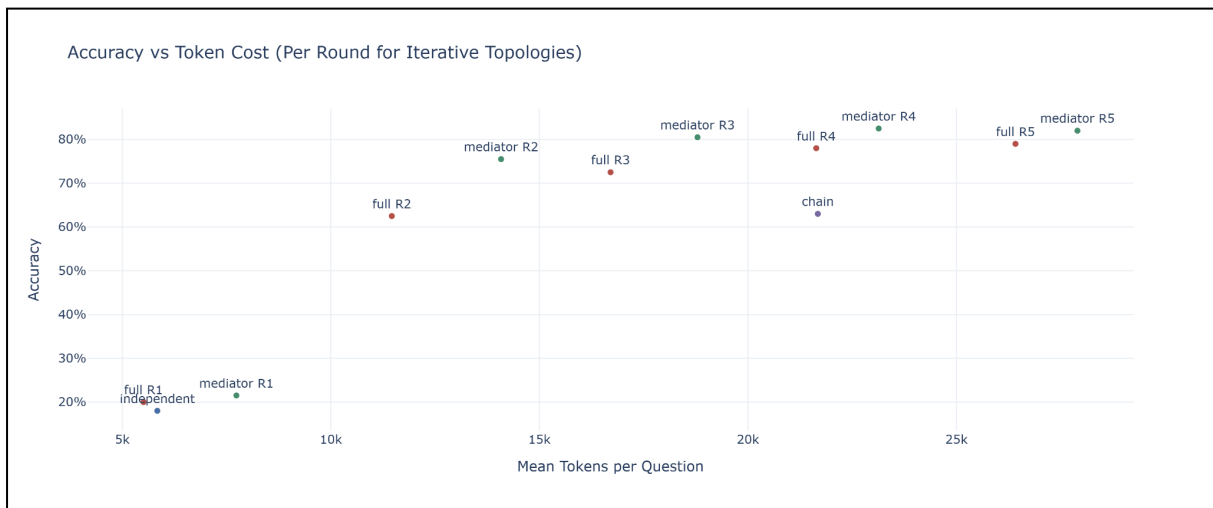


Figure 3

The efficiency-cost based framing of the work cited above motivated the Pareto analysis of Section 5.

In the context of that section, each topology-round combination was evaluated in terms of both accuracy and token cost in order to determine which combinations were not dominated on either of these dimensions.

4. Work Undertaken

This chapter describes the design, implementation, and evaluation methodology of the multi-agent reasoning system. All source code, configuration files, and experimental results are available in the project repository. The system was built as a custom framework rather than relying on existing multi-agent orchestration tools such as AutoGen (Wu et al., 2023), to maintain full control over communication topology constraints.

4.1 Dataset

The experiments make use of the GSM8K benchmark (Cobbe et al., 2021) which contains 8,500 linguistically diverse grade school mathematics problems that need between 2-8 reasoning steps in order to arrive at a correct solution. Each problem will have a unique natural language solution that uses simple arithmetic operations. GSM8K was selected for three reasons: it requires real multi-step reasoning as opposed to just matching patterns, it has one exact numerical answer that can be determined by a computer program, and it is used frequently across the multi-agent debate literature, thus, the results can be compared directly to other work done in this area.

The experiments select a fixed set of 200 problems from the test set using a deterministic seed (seed=0). The size of the sample set is based upon achieving a balance of statistical power and computational practicality. It takes thousands of LLM inference calls to run each topology through 5 rounds, and completing the entire experiment took multiple days on local computing equipment. The same 200 problems are used for comparison of all four topologies allowing for pair-wise statistical comparisons.

4.2 Models and Infrastructure

The four small language models functioned as the reasoning agents:

- Agent 0: Gemma3 4B (Google)
- Agent 1: Phi4-Mini (Microsoft)
- Agent 2: Llama3.2 3B (Meta)
- Agent 3: Qwen2.5 3B Instruct (Alibaba)

All models were run locally through Ollama at 4-bit quantisation, with a shared temperature of 0.4. The choice to use four distinct architectures from different model families rather than four instances of the same model was deliberate: prior work has shown that architectural diversity among debating agents improves reasoning outcomes (Hegazy, 2024). Using small models (3-4B parameters) also tests whether multi-agent collaboration can compensate for individually weak reasoning.

Technical details of each model architecture are documented in their respective reports (Gemma Team, 2025; Microsoft et al. 2025; Grattafiori et al., 2024; Qwen Team, 2024).

Model Summary

Agent	Model	Developer	Parameters	Independent Accuracy
0	Gemma3 4B	Google	4B	9.5%
1	Phi4-Mini	Microsoft	3.8B	23.5%
2	Llama3.2 3B	Meta	3B	7.0%
3	Qwen2.5 3B	Alibaba	3B	19.0%

Figure 4

Summary of the four language models used as reasoning agents. Independent accuracy reflects each agent's individual performance under the independent topology

4.3 Communication Topologies

The four communication topologies were implemented, each defining what information each agent can observe before revising its answer in a given round.

Independent (baseline). Each of the four agents answer the question once, without receiving input from the other agents. The final answer of the group is determined by the majority vote of the four individual answers. The independent communication topology has self-consistent sampling and provides a baseline for comparison to determine if the agents' ability to communicate enhances performance.

Fully connected. Within each round, each agent receives the answers, reasoning, and confidence scores of the other agents. Afterward, each agent revises its answer according to the collective information received. The fully connected communication topology is the most commonly utilized topology in multi-agent debates.

Mediator. In each round, a separate mediator (Agent 0) receives the answers of the agents and generates a neutral summary of the answers submitted by the agents, the number of agents who supported each answer, and the primary reason(s) supporting each answer. Only the summary is communicated to the agents. No agent receives the raw answers submitted by the other agents. Each agent revises its answer after viewing the summary generated by the mediator. The mediator is specifically instructed not to indicate which answer is correct.

Chain. The agents submit their answers sequentially. Agent 0 submits its answer without receiving information from the other agents. Agent 1 receives only the answer submitted by Agent 0 before submitting its answer. Agent 2 receives only the answer submitted by Agent 1 before submitting its answer. The process continues until all agents have submitted their answers. The chain communication topology is the sparsest possible connected communication topology, because each agent receives information from only one other agent. Therefore, there is only one pass through the chain, resulting in the chain being completed in four submissions instead of five rounds.

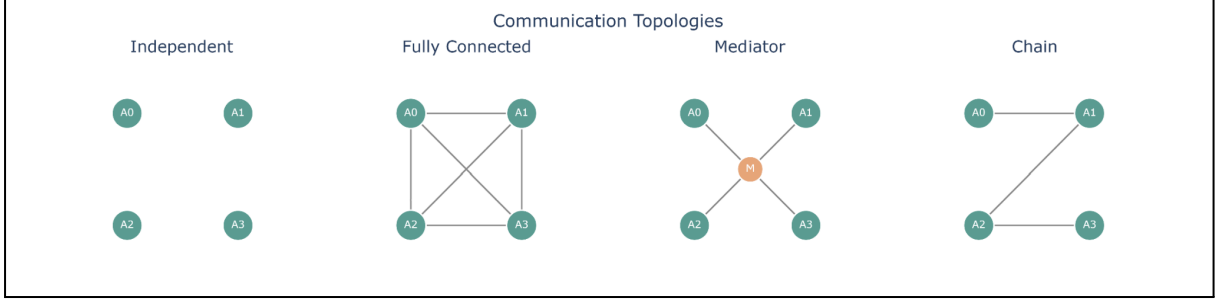


Figure 5

The four communication topologies evaluated in this study. Nodes represent agents (A0-A3); edges represent information flow between rounds

4.4 Agent Response Structure

Each agent submits a structured response with 3 parts to the system in each round:

- **Agent Answer.** A single, numeric value which represents the agent's answer to the problem.
- **Agent Reasoning.** A step-by-step explanation of how they derived the answer.
- **Agent Confidence.** An agent-assigned score (0-100) reflecting the agent's confidence.

Each response is processed automatically. Responses submitted by the agent which can't be processed using the expected format (i.e., responses that have badly formatted text or leave out fields such as the answer field) are labeled as failed parses. As a whole, 8.1% of all responses were considered failed parses. Of those failures, most were attributed to Llama3.2 3B (26-42% failure rate depending on topology). All other models had failure rates under 6%.

4.5 Aggregation

Following the submission of each round, the group's selected answer is determined via majority vote on the 4-agent responses. There is no selection made if there is a tie. The system selects the answer that has the highest average confidence score from the agents who selected that answer. If an agent's response failed to process properly, they would not be eligible to vote in that particular round.

5. Outputs

This chapter contains the experimental outcomes for accuracy, convergence, confidence calibration, cost-efficiency and statistical validation. Each figure shown in this section has been created using the analysis notebook found within the project repository.

5.1 Evaluation Metrics

Experiments are assessed based upon the following evaluation metrics:

Accuracy. The percent of problems that the selected group answer corresponds with the expected correct numeric answer. Accuracy is calculated for the last round for iterative topologies (Fully Connected and Mediator) and for the single pass for Independent and Chain.

Convergence. For iterative topologies, Convergence is quantified in terms of Agreement Rate (Proportion of Agents Providing the Same Answer in a Given Round)

Per-Round Group Accuracy. The metrics above differentiate between agents reaching a common answer (possibly the right answer) and agents providing a common incorrect answer.

Tokens Used: The aggregate number of prompt and completion tokens utilized per question, measured collectively across all rounds.

Tokens per Question (Raw Cost) and Tokens per Correct Answer (Cost Efficiency). They provide comparable metrics to assess the cost required to achieve a particular level of accuracy.

5.2 Statistical Methodology

Statistical validity is needed to verify that differences in accuracy reflect real effects and not just sampling variability since experiments utilize a fixed set of 200 questions. The pipeline to validate includes the following steps:

A Cochran's Q Test is applied as an omnibus test to determine if there is a statistically significant difference in accuracy between the 4 topologies. The Cochran's Q Test is suitable here due to the nature of the data being paired (all 200 questions are present in each condition) and binary (questions are either answered correctly or incorrectly).

McNemar's Tests are then performed pairwise for all 6 pairs of topologies to determine which specific comparisons are significant. A Bonferroni adjustment is applied to ensure that the Family-Wise Error Rate is controlled across the 6 comparisons, thereby setting the alpha level at approximately 0.0083 ($\alpha = .05/6$).

Bootstrap Confidence Intervals (10,000 resamplings) are constructed for both accuracy and token cost at each topology-round combination. The bootstrap intervals measure the uncertainty surrounding each point estimate and are used to create a Pareto Frontier identifying which configurations are neither dominated by others on accuracy nor cost.

5.3 Accuracy Across Topologies

Group accuracy for the four topologies in the last round was as follows; independent 18.0%; Chain 63.0%; Fully Connected 78.0% and Mediator 84.0%. Collaborative improvements in accuracy were observed in each case. Even though Chain was the weakest collaborative topology, its accuracy was more than three times that of the independent model.

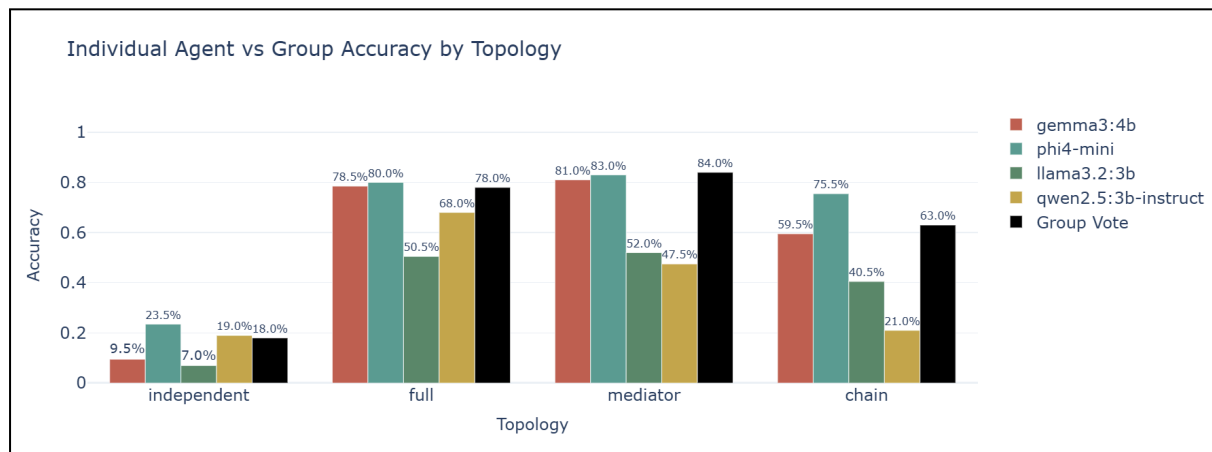


Figure 6

Looking at how well the individual agents performed in this study revealed some interesting differences. When working independently, none of the individual agents could reach better than 23.5% accuracy; when working collaboratively however, using the same models, the best performing individual agent (Phi4-Mini) reached an accuracy level of 83.0% and the worst individual agent (Qwen2.5:3b-instruct) still reached 47.0% accuracy. These results confirm that collaborative problem-solving does not simply provide a way to aggregate each individual's knowledge by voting; but instead provides a method for improving each individual's reasoning by exposing them to different perspectives.

Mediator was the only topology where the group vote (84.0%) was greater than the accuracy of every single individual model. Phi4-Mini, the most accurate individual model, reached an accuracy of 83.0%; in all the other topologies there was at least one model that exceeded the accuracy of the group vote. Therefore, as seen in the fully connected and chain topologies, unrestricted communication among models may dilute the impact of strong individual reasoning. And in the mediator topology, the structured summary of information may be filtering out irrelevant information or "noise" allowing only the relevant information to pass through to the individual models.

5.4 Convergence Dynamics

For the two iterative topologies (fully connected and mediator), both the agreement rate and the group accuracy were measured across the number of rounds.

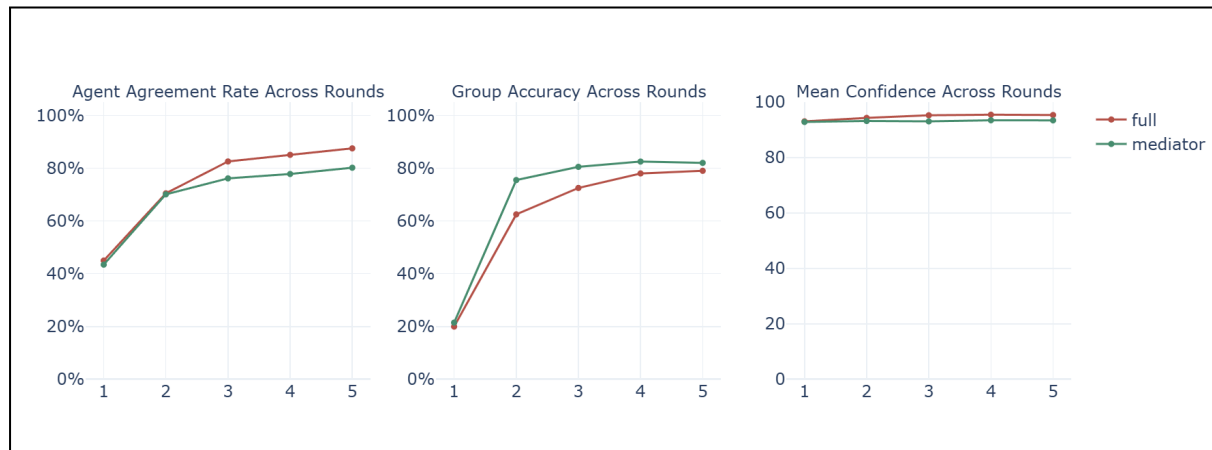


Figure 7

The agreement rates for both topologies began around 45% in round 1 and increased to 85% by round 5. Group accuracy followed a similar pattern and confirmed that the group members were converging toward the correct answers and not simply converging toward an incorrect answer. As seen in the figure above, Mediator reached higher levels of accuracy in fewer rounds. For example, Mediator reached 75.5% accuracy in round 2 whereas fully connected reached 62.5% accuracy in the same round. The structure of Mediator may allow its agents to more quickly identify the most effective reasoning and thus converge faster than fully connected.

In addition to tracking agreement rates and group accuracy, mean confidence also remained relatively constant across the number of rounds and all possible combinations of correct and incorrect answers. The average confidence of the agents remained at ~93-95% regardless of whether their answers were actually correct or not.

5.5 Confidence Calibration

All topologies showed persistent over-confidence. Self-reported confidence ranged from 92.7 percent (independent) to 95.4 percent (fully connected), whereas the average accuracy of the individual's answers ranged from 14.8 percent to 69.2 percent.

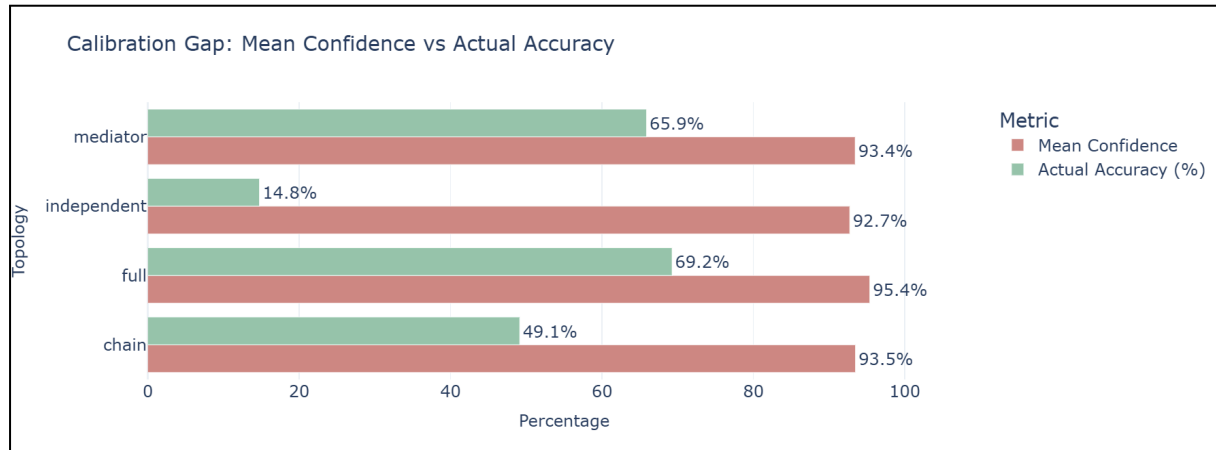


Figure 8

There was a much larger gap between confidence and actual performance under independent reasoning (92.7 percent confidence vs. 14.8 percent) than there was under fully connected (95.4 percent vs. 69.2 percent), although the gap was always large enough so that confidence did not significantly reflect correctness. These results are consistent with Yang et al. (2024) finding that small language models often generate poorly calibrated verbalized confidence estimates, and suggest that at least for models of this size, confidence should not be used as an indicator of the quality of answers in multi-agent systems.

5.6 Answer Flow Between Rounds

To better understand the progression of agent responses during collaboration the flow of responses between rounds was tracked into four categories: Correct-to-Correct; Correct-to-Incorrect; Incorrect-to-Correct; and Incorrect-to-Incorrect.

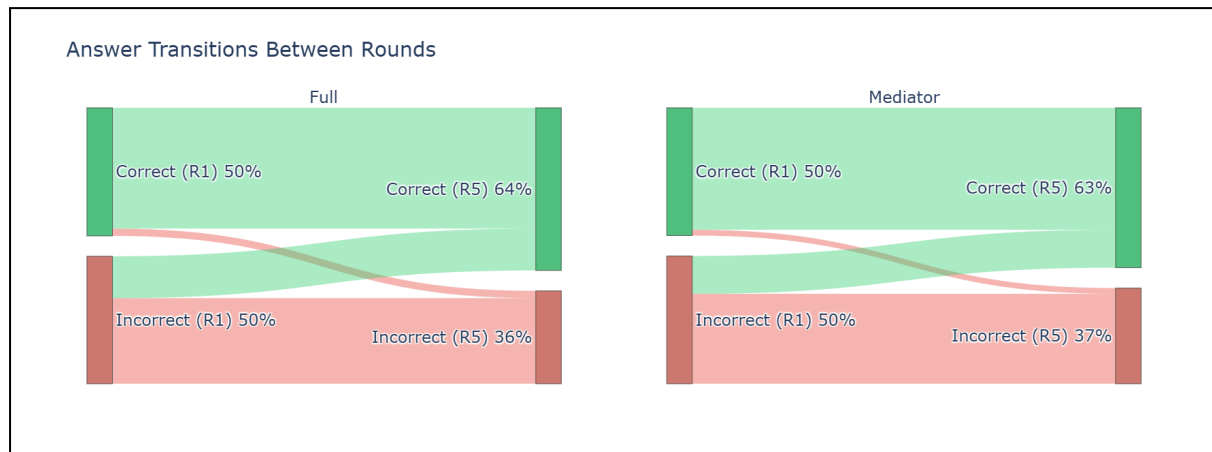


Figure 9

Both topologies demonstrate a significant net correction. The agents begin around 50% correct individually in Round 1 and improve to ~63% correct by Round 5. That's a total improvement of approximately 13 percentage points due to discussion alone. The largest movement is from Incorrect-to-Correct, reinforcing that collaboration allows agents to correct their own errors. There is a much smaller correct-to-incorrect flow present in both topologies (agents can be convinced to give an incorrect response) however this is significantly less than the corrections. Across both topologies, improvements outweigh regressions by more than 6:1. Mediator demonstrates a very thin correct-to-incorrect flow band, indicating that the structured summary does provide some protection from agents being influenced toward incorrect answers.

The final average of the group is higher than the averages for each individual condition because the process of using majority vote to select answers further eliminates the incorrect minority answers.

5.7 Per-Question Error Analysis

All questions did not benefit equally from the collaboration. An analysis of which questions each topology answered and if there was any overlap between their answer sets was performed.

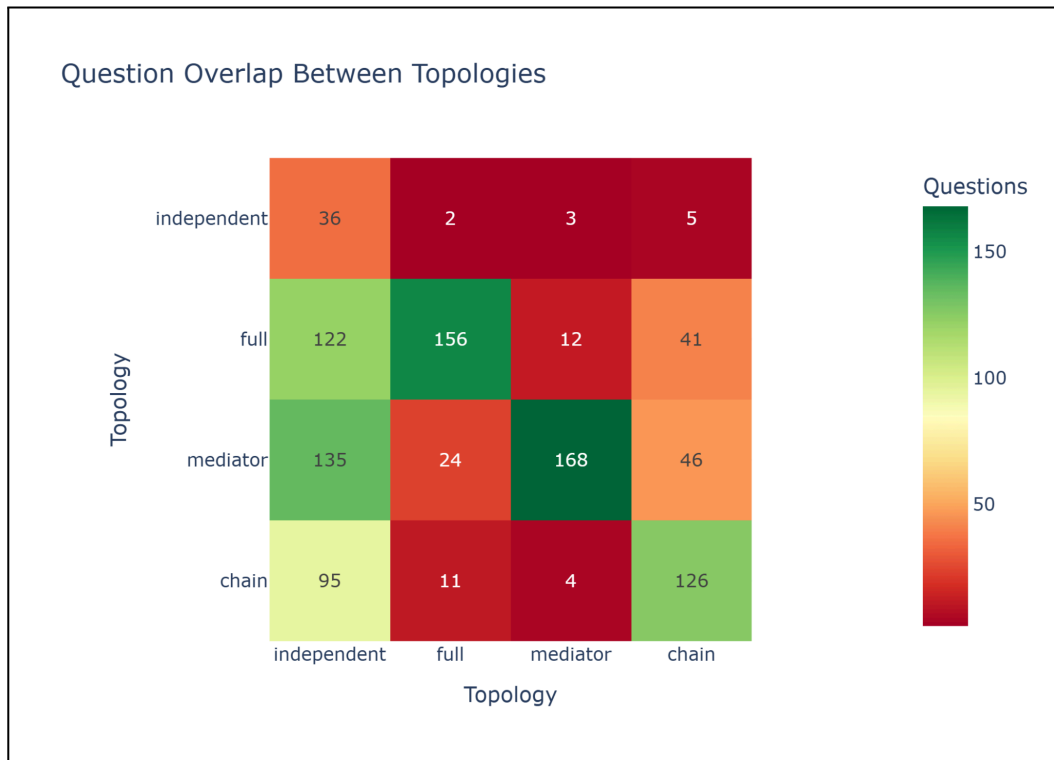


Figure 10

Mediator answered the most questions overall and provided a complete superset of the Independent condition, answering every question that Independent answered. Fully Connected came in second, but nearly all of the questions answered by Fully Connected were also answered by Mediator. Chain answered a reasonable number of questions but only added a handful of new answers. This indicates that Mediator provides the best breadth of answers and that the combination of topologies will provide a small additional benefit beyond Mediator.

5.8 Token Cost and Efficiency

Collaboration costs tokens. The question is whether the accuracy gains justify the expenditure.

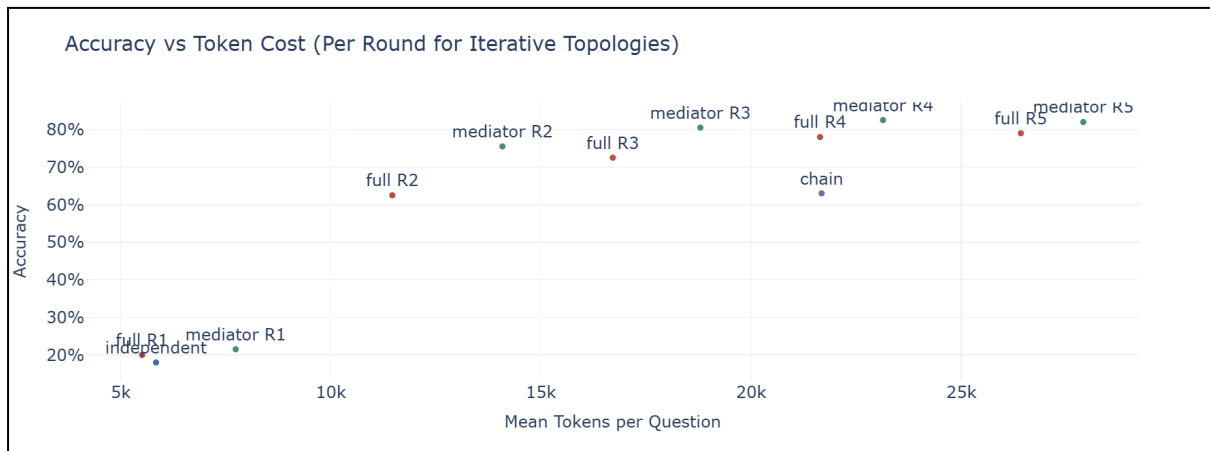


Figure 11

Independent reasoning was the cheapest method at approximately 5,800 tokens per question, but had an accuracy rate of only 18.0%, meaning the cost of achieving a correct answer was about 32,400 tokens. Chain used approximately 21,700 tokens per question to achieve an accuracy rate of 63.0%. Of the iterative methods, Mediator at Round 2 demonstrated a very good balance of accuracy and cost. Mediator demonstrated an accuracy rate of 75.5% at approximately 14,000 tokens per question, providing one of the lowest cost-per-correct-answer values across all methods tested (~18,600 tokens). The addition of more rounds after Round 3 dramatically increased the token usage with little to no increase in accuracy. The cost of going from Round 2 to Round 3 was roughly 5,000 tokens for a 5 percentage point increase in accuracy, while the cost of increasing from Round 3 to Round 4 was another 5,000 tokens for only 2 additional percentage points.

Topology	Accuracy	Tokens/Q	Tokens/Correct	Parse Fail %
Independent	18.0%	5,840	32,445	12.2%
Chain	63.0%	21,676	34,406	7.7%
Mediator: round 1	21.5%	7,736	35,982	8.5%
Mediator: round 2	75.5%	14,081	18,651	8.5%
Mediator: round 3	80.5%	18,790	23,342	8.5%
Mediator: round 4	82.5%	23,135	28,042	8.5%
Mediator: round 5	82.0%	27,900	34,024	8.5%
Full: round 1	20.0%	5,509	27,547	7.3%
Full: round 2	62.5%	11,461	18,338	7.3%
Full: round 3	72.5%	16,707	23,044	7.3%
Full: round 4	78.0%	21,637	27,740	7.3%
Full: round 5	79.0%	26,414	33,436	7.3%

Figure 12

Final accuracy is computed as the majority vote over all four agents' answers and per-round accuracy reported in this summary table differs as it's computed at each round independently.

5.9 Statistical Validation

To confirm that the observed accuracy differences are not artefacts of the 200-question sample, we applied a three-step statistical testing pipeline comparing the four topologies.

Cochran's Q Test. Determine if there is a significant difference in the degree of accuracy between the four topologies. Cochran's Q Test determined that there is a significant difference between the four ($Q = 32.12$; $P = 1.06 \times 10^{-7}$), indicating that at least one topology has a significantly higher degree of accuracy than the others.

Pairwise McNemar Tests With Bonferroni Correction. Determine which topologies have a significantly higher degree of accuracy than the others. The McNemar tests compared the degree of accuracy of each of the collaborative topologies to the degree of accuracy of the independent model. Each of the collaborative topologies had a significantly higher degree of accuracy than the independent ($P < 0.001$ for each). Additionally, the McNemar tests compared the degree of accuracy of the mediator and fully connected topologies to the degree of accuracy of the chain topology. Both mediator and fully connected had a significantly higher degree of accuracy than chain ($P < 0.001$ for each topology). The McNemar tests also compared the degree of accuracy of the mediator and fully connected. The comparison between the mediator and fully connected was not significant after applying the Bonferroni correction ($P = 0.40$), indicating that although there is no significant difference in the degree of accuracy between the mediator and fully connected, they are functionally equivalent.

Pair	A only Correct	B only Correct	p (raw)	p (Bonferroni)	Sig
independent vs chain	5	95	5.5847e-19	3.3508e-18	***
independent vs full	2	122	1.1771e-26	7.0626e-26	***
independent vs mediator	3	135	7.0441e-29	4.2264e-28	***
chain vs full	11	41	5.7810e-05	3.4686e-04	***
chain vs mediator	4	46	6.7000e-09	4.0200e-08	***
full vs mediator	12	24	6.6753e-02	4.0052e-01	ns

Figure 13

Bootstrap Confidence Intervals. To verify the stability of the results obtained from the McNemar tests, bootstrap confidence intervals (10,000 resamples) were calculated. The bootstrap confidence intervals verified the results obtained from the McNemar tests.

Pareto Frontier Analysis: In addition to determining which topology has a significantly higher degree of accuracy, the Pareto frontier analysis determined whether or not the mediator topology is the most efficient topology at achieving high levels of accuracy. At any given level of token budget, the

mediator achieved the highest degree of accuracy. Additionally, the Pareto frontier analysis determined whether or not the mediator was superior to the chain and fully connected. The chain and fully connected topologies were consistently inferior to the mediator, indicating that a mediator configuration always existed that was cheaper and had a higher degree of accuracy than either of the chain or fully connected configurations.

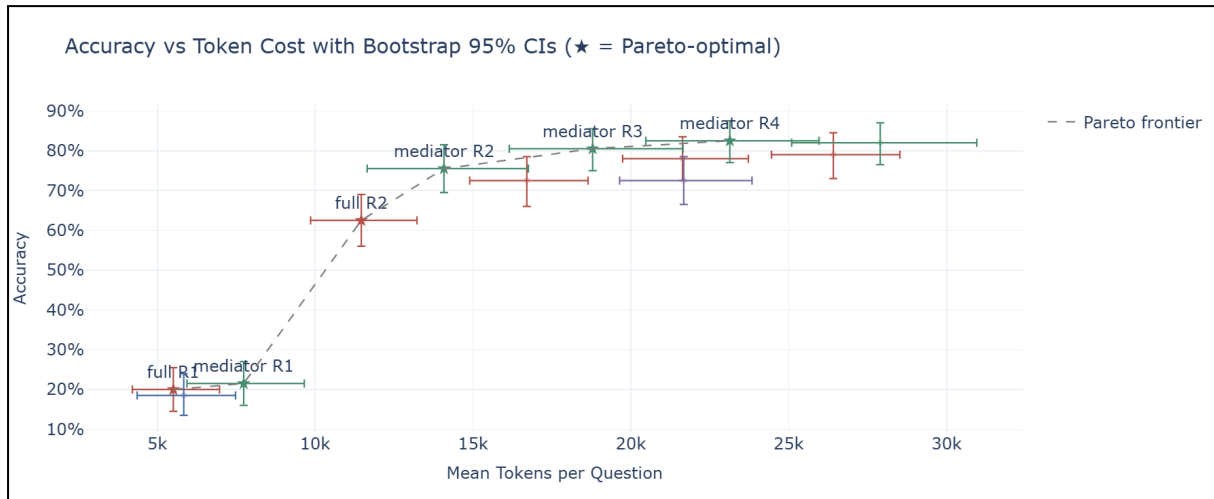


Figure 14

5.10 Baseline Comparison: Self-Refine

In order to isolate the effect of multi-agent communication on the accuracy of the model, a comparison was made between mediator (mediator R2) and the self-refine model. Although both topologies involved multiple rounds of reasoning, the self-refine model did not expose the agent to the diverse perspectives available from the other model architectures.

The self-refine model achieved an accuracy of 53.0% at an average cost of 3200 tokens per question. This represented a +35.0 percentage point increase in accuracy when compared to the independent voting model (which had an accuracy of 18.0%). Therefore, self-refine demonstrates that multiple rounds of self-criticism can be beneficial for a single model's ability to reason accurately. However, the self-refine model fell well short of the accuracy achieved by the collaborative topologies. Mediator R2 achieved an accuracy of 75.5% at round 2. This represented a +22.5 percentage point increase in accuracy over the self-refine model.

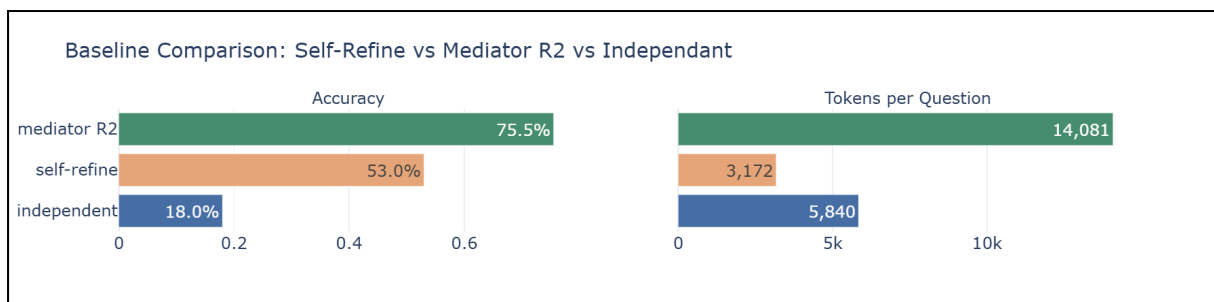


Figure 15

This comparison clearly shows that a model critiquing its own work cannot overcome the limitations of its own blind spots. It requires additional input from agents with different reasoning patterns to identify and correct errors that self-reflection alone fails to detect.

The trade-off in terms of the number of tokens required for self-refine versus mediator R2 is also instructive. Self-refine requires approximately 3,200 tokens per question, whereas mediator R2 requires approximately 14,000 tokens per question. Therefore, the mediator R2 model provides a 4.4 fold increase in cost relative to the self-refine model for a 22.5% point increase in accuracy. For applications where correctness is paramount, this may represent a favorable trade-off.

Mediator R2 represents the most practical compromise in terms of the accuracy and token efficiency ratio. As previously demonstrated by the Pareto frontier analysis, it represents the optimal compromise between the two competing metrics.

6. Future Work and Limitations

6.1 Future Work

There are several ways to expand upon the findings of this study notably with larger models. These experiments have been carried out with 3-4B parameter models, which run at 4-bit quantization. It is not clear if the topology results shown here will persist at a larger scale. For example, will a mediator topology continue to be superior to fully connected at scales of 7B or 14B, given that each agent will be significantly better at reasoning individually? Topology effects may decrease as individual agent capabilities improve. Alternatively, it is possible that stronger agents will be able to take advantage of structured communication to an even greater extent.

Wider Task Domains. All the experiments were carried out with GSM8K which is a mathematical reasoning benchmark with one correct numerical answer. Communication topology may interact with tasks that allow for open-ended or subjectively correct answers (such as code generation, creative writing, multi-step planning) in different ways. Specifically, a mediator's neutral summary function may be less well-suited for tasks with no single correct answer to converge on.

Topology Selection Based Upon Question Difficulty. The current system always has a fixed topology for every question. A system that can dynamically select the topology based upon question difficulty, or upon the level of disagreement among agents may be able to combine the best features of each structure. For example, agents could reason independently for easier questions and use mediation for harder questions.

Early Termination of Rounds. The current system uses a fixed number of rounds (5) for all iterative topologies. However, the convergence analysis indicates that most of the accuracy gains occur in rounds 2 and 3. Implementing an early-stopping strategy to terminate the conversation when the agents agree (or when accuracy gain falls below a certain threshold), should significantly reduce the number of tokens required to achieve the same level of performance.

Better Confidence Calibration. The confidence scores generated by the small models were not informative. Therefore, future research may investigate whether the confidence scores can be improved through different prompting strategies, for example, asking the agents for probability estimates rather than confidence percentages or using the consistency of their responses across rounds as a surrogate for confidence.

More Agents. This study was carried out with four agents. Further investigation into whether increasing the number of agents (for example 6-8) improves the accuracy or simply increases the cost, and how the optimal topology varies with group size will assist in the generalizing of the results.

6.2 Limitations

- Small single task (only GSM8K); the findings likely do not generalize to open-ended or non-mathematical tasks.
- The sample used is small in relation to the total number of questions sampled (200/8500) although the results have been statistically validated, a larger sample would strengthen confidence in the findings.
- Only small models were used (3-4B), unclear from my own work and method whether topology effects hold at 7B+ scales.
- Single hardware configuration (one RTX 4070 Ti, sequential Ollama inference), no parallelism, which constrained sample size and round count.
- Each topology tested had only one trial conducted; since Llama3.2 operates stochastically at $T=0.4$, we can expect slight variations on the results from one trial to another.
- Unhelpful Confidence Scores, the self-report confidence score of 0-100 provided no assistance when aggregating data or making decisions regarding early termination.
- Llama3.2 3B high parse failure rate (26-42%), this agent's unreliability may have disproportionately affected some topologies more than others.
- Mediator uses a fixed agent (Gemma3 4B / Agent0), the choice of mediator model was not varied, so mediator performance may partly reflect that specific model's summarisation ability rather than the topology alone.

7. Conclusion

This research project investigated whether communication topology affects the degree of accuracy of reasoning and rate of convergence in Multi-Agent Large Language Model (LLM) Systems. Our experimental results show a strong affirmation.

We tested four types of communication topology; independent, fully connected, mediator and chain across 200 GSM8K mathematical reasoning tasks using four small size LLM. Independent reasoning produced an 18.0% group accuracy. All collaborative topologies showed an increase on this baseline with chain producing 63.0%, fully connected producing 78.0% and mediator producing 84.0%. We used Cochran's Q test and McNemar's pairwise tests to confirm these differences were statistically significant.

The mediator topology performed as the best configuration across a number of metrics. It was the most accurate, it converged faster than fully connected (producing 75.5% by round 2 vs 62.5%) and it dominated the Pareto frontier from round 2 onwards, indicating that for any given token budget, no other topology was able to produce higher accuracy. Also, the mediator was the only topology where majority voting outperformed each of the individual agents at all times, suggesting that the mediator's structured summary filtering is able to filter out noise much more effectively than raw peer-exposure.

In the supplementary self-refine experiment we found that multi-agent collaboration has advantages beyond those offered by iteratively refining one's own critique. Phi4-Mini self-refined over five rounds to achieve 53.0% accuracy, an improvement of 35 percentage points over the independent group baseline, yet still 22.5 percentage points behind mediator at round two. This difference indicates the value of having diverse external perspectives, since a model critiquing its own reasoning is unable to discover errors requiring a fundamentally different type of reasoning approach to expose them.

We observed that Confidence Calibration was not reliable under any of our tested conditions, as Mean Reported Confidence exceeded 93% regardless of correctness. This result is consistent with prior research on verbalized confidence in small models and suggests that confidence values should not be used as a basis for aggregation or early termination in systems of this type.

Our findings support the view that Communication Topology represents a critical design factor when designing Multi-Agent Reasoning Systems. Practitioners developing such systems using small open-source models have a clear recommendation based upon the results of this study: structured mediation produces a greater accuracy-to-cost ratio than either unstructured discussion or sequential communication. Finally, the results of this study motivate additional research into adaptive topology selection, larger model scales and broader task domains.

8. References

- Chebotarev, P. (2010). Comments on ‘Consensus and cooperation in networked multi-agent systems’. *Proceedings of the IEEE*, 98(7), 1353–1354.
<https://doi.org/10.1109/JPROC.2010.2049911>
- Chuang, Y.-S., Goyal, A., Harlalka, N., Suresh, S., Hawkins, R., Yang, S., Shah, D., Hu, J., & Rogers, T. T. (2024). Simulating Opinion Dynamics with Networks of LLM-based Agents (arXiv:2311.09618). arXiv. <https://doi.org/10.48550/arXiv.2311.09618>
- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., & Schulman, J. (2021). Training Verifiers to Solve Math Word Problems (arXiv:2110.14168). arXiv. <https://doi.org/10.48550/arXiv.2110.14168>
- Du, Y., Li, S., Torralba, A., Tenenbaum, J. B., & Mordatch, I. (2023). Improving Factuality and Reasoning in Language Models through Multiagent Debate (arXiv:2305.14325). arXiv. <https://doi.org/10.48550/arXiv.2305.14325>
- Estornell, A., & Liu, Y. (n.d.). Multi-LLM Debate: Framework, Principals, and Interventions.
- Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Vaughan, A., Yang, A., Fan, A., Goyal, A., Hartshorn, A., Yang, A., Mitra, A., Sravankumar, A., Korenev, A., Hinsvark, A., ... Ma, Z. (2024). The Llama 3 Herd of Models (arXiv:2407.21783). arXiv. <https://doi.org/10.48550/arXiv.2407.21783>
- Huang, J., Chen, X., Mishra, S., Zheng, H. S., Yu, A. W., Song, X., & Zhou, D. (2024). Large Language Models Cannot Self-Correct Reasoning Yet (arXiv:2310.01798). arXiv. <https://doi.org/10.48550/arXiv.2310.01798>
- Kotelanski, M., Gallo, R., Nayak, A., & Savage, T. (n.d.). Methods to Estimate Large Language Model Confidence.
- Li, Y., Du, Y., Zhang, J., Hou, L., Grabowski, P., Li, Y., & Ie, E. (2024). Improving Multi-Agent Debate with Sparse Communication Topology (arXiv:2406.11776). arXiv. <https://doi.org/10.48550/arXiv.2406.11776>

- Liang, T., He, Z., Jiao, W., Wang, X., Wang, Y., Wang, R., Yang, Y., Shi, S., & Tu, Z. (2024). Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate (arXiv:2305.19118). arXiv. <https://doi.org/10.48550/arXiv.2305.19118>
- Madaan, A., Tandon, N., Gupta, P., Hallinan, S., Gao, L., Wiegreffe, S., Alon, U., Dziri, N., Prabhumoye, S., Yang, Y., Gupta, S., Majumder, B. P., Hermann, K., Welleck, S., Yazdanbakhsh, A., & Clark, P. (2023). Self-Refine: Iterative Refinement with Self-Feedback (arXiv:2303.17651). arXiv. <https://doi.org/10.48550/arXiv.2303.17651>
- Microsoft, Abouelenin, A., Ashfaq, A., Atkinson, A., Awadalla, H., Bach, N., Bao, J., Benhaim, A., Cai, M., Chaudhary, V., Chen, C., Chen, D., Chen, D., Chen, J., Chen, W., Chen, Y.-C., Chen, Y., Dai, Q., Dai, X., ... Zhou, X. (2025). Phi-4-Mini Technical Report: Compact yet Powerful Multimodal Language Models via Mixture-of-LoRAs (arXiv:2503.01743). arXiv. <https://doi.org/10.48550/arXiv.2503.01743>
- Qian, C., Xie, Z., Wang, Y., Liu, W., Zhu, K., Xia, H., Dang, Y., Du, Z., Chen, W., Yang, C., Liu, Z., & Sun, M. (2025). Scaling Large Language Model-based Multi-Agent Collaboration (arXiv:2406.07155). arXiv. <https://doi.org/10.48550/arXiv.2406.07155>
- Qwen, Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Li, C., Liu, D., Huang, F., Wei, H., Lin, H., Yang, J., Tu, J., Zhang, J., Yang, J., Yang, J., Zhou, J., Lin, J., ... Qiu, Z. (2025). Qwen2.5 Technical Report (arXiv:2412.15115). arXiv. <https://doi.org/10.48550/arXiv.2412.15115>
- Smit, A., Duckworth, P., Grinsztajn, N., Barrett, T. D., & Pretorius, A. (2024). Should we be going MAD? A Look at Multi-Agent Debate Strategies for LLMs (arXiv:2311.17371). arXiv. <https://doi.org/10.48550/arXiv.2311.17371>
- Snell, C., Lee, J., Xu, K., & Kumar, A. (2024). Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters (arXiv:2408.03314). arXiv. <https://doi.org/10.48550/arXiv.2408.03314>
- Team, G., Kamath, A., Ferret, J., Pathak, S., Vieillard, N., Merhej, R., Perrin, S., Matejovicova, T., Ramé, A., Rivière, M., Rouillard, L., Mesnard, T., Cideron, G., Grill, J., Ramos, S.,

- Yvinec, E., Casbon, M., Pot, E., Penchev, I., ... Hussenot, L. (2025). Gemma 3 Technical Report (arXiv:2503.19786). arXiv. <https://doi.org/10.48550/arXiv.2503.19786>
- Wang, J., Wang, J., Athiwaratkun, B., Zhang, C., & Zou, J. (2024). Mixture-of-Agents Enhances Large Language Model Capabilities (arXiv:2406.04692). arXiv. <https://doi.org/10.48550/arXiv.2406.04692>
- Wang, Q., Wang, Z., Su, Y., Tong, H., & Song, Y. (2024). Rethinking the Bounds of LLM Reasoning: Are Multi-Agent Discussions the Key? (arXiv:2402.18272). arXiv. <https://doi.org/10.48550/arXiv.2402.18272>
- Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., & Zhou, D. (2023). Self-Consistency Improves Chain of Thought Reasoning in Language Models (arXiv:2203.11171). arXiv. <https://doi.org/10.48550/arXiv.2203.11171>
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., & Zhou, D. (2023). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models (arXiv:2201.11903). arXiv. <https://doi.org/10.48550/arXiv.2201.11903>
- Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., Awadallah, A. H., White, R. W., Burger, D., & Wang, C. (2023). AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation (arXiv:2308.08155). arXiv. <https://doi.org/10.48550/arXiv.2308.08155>
- Yang, D., Tsai, Y.-H. H., & Yamada, M. (2024). On Verbalized Confidence Scores for LLMs (arXiv:2412.14737). arXiv. <https://doi.org/10.48550/arXiv.2412.14737>
- Zhang, G., Yue, Y., Sun, X., Wan, G., Yu, M., Fang, J., Wang, K., Chen, T., & Cheng, D. (2025). G-Designer: Architecting Multi-agent Communication Topologies via Graph Neural Networks (arXiv:2410.11782). arXiv. <https://doi.org/10.48550/arXiv.2410.11782>
- Hegazy, M. (2024). Diversity of thought elicits stronger reasoning capabilities in multi-agent debate frameworks. arXiv preprint arXiv:2410.12853.